Fisheries Information System

National Observer Program

FY 2025 Project Proposal

Project: Building a curated Alaska Fisheries Science Center genetic sample repository to facilitate long-term use of molecular information for fisheries management.

Principal Investigator: Diana Baetscher, Patrick Barry

Area of Interest: FIN Development

Date: 06/24/2024

National Marine Fisheries Service

Contact: FIS.PMO@noaa.gov

Proposal must be approved by the Regional Administrator/Deputy, Science Center Director/Deputy, Headquarters Office Director/Deputy, or equivalent. Multiple submissions from one organization should be prioritized. The FIS Program Management Office will request approval and prioritization of proposals by the approvers prior to proposal review.

Building a curated Alaska Fisheries Science Center genetic sample repository to facilitate long-term use of molecular information for fisheries management.

Project Description

1. Amount Funded

2. Organization

Alaska Fisheries Science Center

3. Jointly Submitted by Multiple Regions

Y

4. If you have previously received funding from this RFP, are the quarterly reports up-to-date and have you submitted a final report for all completed projects?

NA

5. Project Goal

This project's primary goal is to create a database and interface for the Auke Bay Laboratories inventory of genetic samples to facilitate the organization, access, and reporting of genetic data to inform fisheries management.

6. Executive Summary

The Alaska Fisheries Science Center Genetics Program (AFSC GP) conducts research on Pacific salmon, groundfish, and forage fish species to comply with regional, national, and international agreements and treaties for managing marine resources. Much of this work relies on the collection of samples, both fishery dependent and independent, to evaluate the existence and extent of population genetic structure in a variety of species. Currently, the genetics program manages a database of these samples, associated collection information, and laboratory data in FileMaker software. Filemaker is a relational database application with a graphical user interface. Recent limitations with the current database highlight the need for better records management. This project's primary goal is to work with the Alaska Fisheries Information Network (AKFIN) to develop a database model and repository for AFSC GP's inventory of genetic samples. Additionally, a data management and upload tool in Oracle APEX will be created to interface with these data. These tables will be used for storing

sample information including species, tissue type, collection information, and laboratory processing data. Effective organization of these samples will aid in both the evaluation of population genetic trends across a diverse array of taxa as well as the development of molecular resources for fisheries management. This project would bring consistency to AFSC GP's records management, as current salmon bycatch analyses rely heavily on AKFIN's database.

7. Describe how project supports FIS/ET/CSP goals and objectives

This project will support the creation of a new more robust database for AFSC GP and will enhance the lab's ability to access useful sample metadata. Whenever possible, the lab revisits archived tissue samples to design new experiments to evaluate theories and develop molecular resources that can inform fisheries management. A functional and durable database is foundational to this work and will enhance the curation, modification, and access to sample data. Integration of AFSC GPs sample collection into the AKFIN ecosystem will lead to efficiencies and reproducibility in AFSC GP workflows.

While the transition to the legacy FileMaker database moved away from individual researchers managing data within excel files on a shared network drive, it was developed by research geneticists with varying data types, degrees of investment, and database experience. The creation and management of a new repository by database professionals with input from the AFSC GP would yield a product with improved architecture and durability. The implementation of data field validation checks would prevent records from having nonsensical values and would standardize field formatting which would lead to greater data integrity. The improved schema design would include staging, storage and reporting environments.

Additionally, a major outcome of this project would improve accessibility to AFSC's data. The legacy database implemented in FileMaker software is run off of a virtual machine managed by NOAA in Seattle. Two users cannot access the database at the same time because the software requires the use of a graphical user interface with a limited number of licenses. The new database will not be limited in this way, and will offer users with different levels of coding abilities to query the database directly with SQL queries, or through R within existing workflows, or through an Oracle APEX application. Additionally, the project will allow for improved sharing among programs by creating reports tailored to certain collaborative projects.

8. Describe the problem the project is addressing

This project's primary goal is to transition the Auke Bay Laboratories inventory of genetic samples and data from a legacy database (FileMaker) to a newly developed database to improve data organization, accessibility, sharing, integrity and consistency. The current database solution lacks rigorous structure and management. This has led to the need for AFSC geneticists to reanalyze multiple datasets which hinders productivity, delays the timely delivery of results, and has substantial computational costs. If this problem is not addressed, AFSC GP will continue to clean up and use its current FileMaker database which is characterized by data consistency issues and access limitations. The major benefit and intended outcomes of this project are to define table structures, clean existing

records, create a new database managed by AKFIN with application development to query and upload records. This would increase access to archived sample metadata, consistency with salmon bycatch data already managed by AKFIN, and provide a foundation for future data/collections. While the proposals scope is specific to the AFSC GP, their legacy database issues are not unique, and the outcomes of this project including the database structure and applications could be applied across regional genetic programs and FINs.

9. Describe the objectives of the project

The goal of this project is to develop a functional relational database management application to store, validate, and report on the genetic sample collection at the AFSC GP. Milestones for the project include designing tables and determining data fields, processing current database records to adhere to new table formatting, and developing the new database and application. The end-point of this project is a functional AFSC genetics database that enhances organization and accessibility of data, increases reproducibility of data analysis, and facilitates data sharing with partner labs (regional science centers and universities).

10. Describe how you are going to achieve those objectives

The AFSC GP will be responsible for assessing the current database format and developing new table structure, determining what legacy fields (~70) will be migrated to the new database and what data type (string, numeric, etc.) they will be stored as. The AFSC GP will check the ~40,000 records in the current database and validate fields for conformance to the newly defined database structure. AKFIN will be responsible for managing contractors that will be developing the database and application. AKFIN personnel would work with AFSC geneticists to create a database model and repository for their sample collection. Additionally, ORACLE data management and upload tools would be developed,and a download and reporting dashboard in Oracle APEX would be implemented. This management and reporting tool will build off a previously developed, FIS-funded application that supports the Alaska Fisheries Science Center Ecosystem and Socioeconomic Profiles (ESP) and Ecosystem Status Reports (ESR).

11. Supporting evidence that project is a priority for the region

FIN Charter/Strategic Plan

12. Explanation

This project aligns with the AKFIN mission To host, integrate, manage, and distribute high-quality data to support Pacific marine fisheries science and management through collaboration with our partners and stakeholders. Additionally, integrating the AFSC GP data into the AKFIN database makes these data "cloud ready", as the FINs will likely migrate databases to cloud hosting in the near future.

13. Description of fishery

The genetics program's work informs management of the Bering Sea, Aleutian Islands, and Gulf of Alaska pollock trawl fisheries, as well as fisheries for rockfish, Pacific cod, pollock, sablefish, and king crab in the Bering Sea, Aleutian Islands, and Gulf of Alaska. Tissue samples are obtained from both fishery independent and fishery dependent sources (e.g., AFSC bottom trawl surveys, AFSC Observer Program, ADFG, NWFSC Genetics Program, and others). Genetic data generated from these tissue samples are then used for stock-wide analyses including characterizing population genetic structure for use in stock assessments that are provided to the fisheries management councils. For example, genetic data from Pacific cod are being used to understand stock boundaries and individual movement across boundaries. Models incorporating this movement are being explored by assessment scientists. Another project uses genetic species identification of tissue samples from blackspotted and rougheye rockfishes collected by the Electronic Monitoring (EM) Research and Development team to evaluate the efficacy of artificial intelligence (AI) for species identification. Ensuring that accurate metadata are preserved for these data products is paramount.

14. Describe your metadata documentation and data documentation plan

Data documentation is a core objective of this proposed project. Internal documentation on the preprocessing of data for transition from FileMaker to AKFIN database structure will be compiled along with methodologies and programming code for the database and application (e.g., data model, lookup tables, ETL code, and validation procedures). These documents will be stored in AKFIN's documentation repository (Confluence) and made available upon request. External documentation which includes table and field comments, as well as a user guide will be available to end users (AFSC GP and potentially collaborators). If and when the database structure is modified all documentation will be updated.

15. Will the finished product be supported and maintained by IT personnel inside or outside of an IT department (e.g., new web application which will be deployed on NOAAs network)?

The AFSC genetics repository, management application, and reports suite within the AFKIN database will be maintained by AKFIN post-development.

16. Describe your transition plan for this project, including a specific description of the product being created, any tasks needed after this project is completed to get the work into operations, what funding and resources will be needed, where that funding will come from, and a description of roles and responsibilities

The product being created is a repository and application for functional storage and retrieval of data associated with genetic samples. After this project is completed, the newly developed database, applications, and documentation will be maintained by AKFIN with existing funds.

17. If partially funded, would this project be able to achieve objectives? If so, in what way?

This project would not be able to achieve the objectives with partial funding.

18. Describe any internal or external matching or in-kind funding

AKFIN and AFSC GP will provide in kind support through staff time, servers, and maintenance of the database and APEX report procedures. Some work has already begun to perform QA/QC on existing data by AFSC geneticists.

19. If funded, where should funds be sent (BOPed) to? Please name the FMC and budget contact if known.

PSMFC/AKFIN - Ngu Castro

Project Leadership

Project Leader and Members

First Name	Last Name	Title	Role	Affiliation	Email	Phone 1	Phone 2
Rob	Ames		Team Member	AKFIN	RAmes@psmfc.org		
Diana	Baetscher	Research Geneticist	Principal Investigator	NOAA	diana.baetscher@noaa.gov	3233721737	
Patrick	Barry	Research Geneticist	Principal Investigator	NOAA	patrick.barry@noaa.gov	5105206994	
Matt	Callahan		Team Member	AKFIN	matt.callahan@noaa.gov		
Natasha	Howe	Data Analyst	Team Member	PSMFC (NOAA Affiliate)	natasha.howe@noaa.gov		
Jamie	Musbach	Genetics Lab Technician	Team Member	PSMFC (NOAA Affiliate)	jamie.musbach@noaa.gov		

Project Estimates

Project Schedule

Item #	Schedule Description	Prerequisite	Start Date	Finish Date	Milestone
1	Receive funding and scheduled bimonthly meetings between AFSC GP, AKFIN,		07/01/2025	07/31/2025	N

	and contract personnel.				
2	Define database tables using existing FileMaker database structure, define fields types.		07/01/2025	08/16/2024	Y
3	Pre-process all existing data in FileMaker database to adhere to data type definitions.	Item 2 - define tables and data types.	08/18/2025	09/19/2025	Y
4	Database development - develop data model, lookup tables, ETL code, validation procedures, setup views for direct access.	Items 2,3 - Existing data must be preprocessed and new database tables/fields defined	09/22/2025	12/19/2025	Y
5	Apex Application Development	Item 4 - database development	01/05/2026	06/30/2026	Y
6	Data documentation - Develop documentation on database and reporting tools as well as a user guide for AFSC GP	Items 3, 4, and 5	07/01/2025	06/30/2026	N

Cost Estimates

Cost Name	Cost Description	Cost Amount
Consultants/Contracts	Application development - setup genetics module in the AFSC DM application or optionally copy app to support geneticists, create & implement authentication, implement genetics data for upload, setup application dashboard, create uploader tool, develop admin functions, application reports/export page, data view and update forms, testing, testing and validation.	$25,750.00
Consultants/Contracts	Database development - develop data model, lookup tables, ETL code, validation procedures, setup views for direct access	$12,750.00
Consultants/Contracts	Develop reporting tools - develop answers repository and dashboard reports.	$8,500.00
Consultants/Contracts	Documentation - technical and end-user documentation.	$5,500.00
	TOTAL COST	**$52,500.00**

Risk

Project Risk

Risk Description	Risk Impact	Risk Probability	Risk Mitigation Approach
Failure in foresight of useful metadata to include with samples	Database restructuring (adding fields) - lack of information for some samples.	Low	Evaluate current database fields for utility, confirm with other regional labs about suitability of final datafields.
Failure to QA/QC all current records within the given time frame	Delayed implementation of full database	Low	Distribute workload among multiple members of the AFSCs genetics program. Create a work plan to outline goals, assign tasks, and implement a timeline. ABL GP has already begun the QC/QA of the existing database.
Failure to develop database	Project would not be successful and AFSC GP would maintain current database	Low	This is very unlikely to occur as AKFIN has extensive experience working with the AFSC GP on salmon bycatch. The two groups currently have monthly meetings to detail progress on progress and outline tasks. A similar approach will be taken for this project.
Failure to develop application for data management, upload and download tools, and reporting dashboard.	AFSC would access the database directly.	Low	Again, this is unlikely as this aspect of the project is able to build off a previously developed, FIS-funded application that supports the Alaska Fisheries Science Center Ecosystem and Socioeconomic Profiles (ESP) and Ecosystem Status Reports (ESR).

Supporting Documents